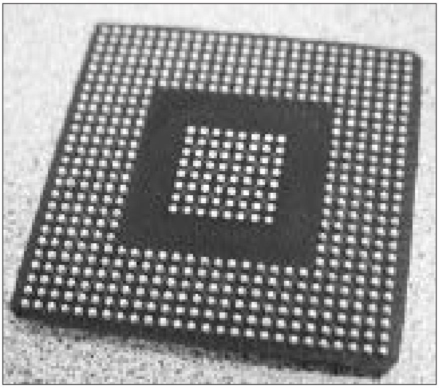
material. The advantage is that unlike pins, the conductors are not easily bent. Unfortunately, BGA CPUs are soldered right onto the motherboard, and users cannot replace them. This form of packaging is seen in some embedded CPUs like those from VIA.



Screenshot 16: BGA chip.jpeg

1.3.3 Cache memory

All CPUs presently made have two components in their die - the processing core and some memory. The memory is used to store frequently-used data so that the CPU doesn't have to wait for the data to be fetched from other, slower storage areas like the system RAM or hard disk. This temporary storage is referred to as the cache. Unlike system RAM, cache is made of more expensive Static RAM, which does not require refreshing. There can be multiple caches per CPU. The cache that is closest to the CPU core is called the Level 1 cache, and is most frequently accessed by the CPU. The subordinate cache, called the Level 2 cache, is approached only if the data is not available in the L1 cache. Some high-end CPUs also sport a Level 3 cache. Cache sizes tend to increase with their Levels, with L1 caches being smaller than L2. In current CPUs, the L1 cache tops out at 128 KB per core, while the L2 cache tops out at 8 MB per core.

1.3.4 Cool 'n' Quiet / Speed Step

These refer to the power management schemes used by AMD and Intel respectively. This allows the CPU to conserve energy by altering its speed according to the processing load. At slower speeds, the power consumption is decreased.

1.3.5 Die size

A Die refers to the block of silica that contains the core logic of a